



RoboFest Gujarat 5.0 (Senior Category) Ajay Kumar Garg Engineering College



Aerial Robotics for Minefield Navigation



Core Technical Contributions

01

Leader–Swarm–Follower role-based drone architecture

02

Gesture-based human intent input (hands-free control)

03

Distributed sensing with centralized planning

04

Human-safe navigation with active guidance

05

Planned thermal-assisted mine detection (FLIR Lepton 3.5)

System

Architecture

Workflow

1

Human Hand
Gesture

2

Leader Drone (RPi 5)
Gesture Processing +
Global Planning

3

Swarm Drones (RPi 4)
[Area Scan + Mine
Detection]

4

Central Map
Update

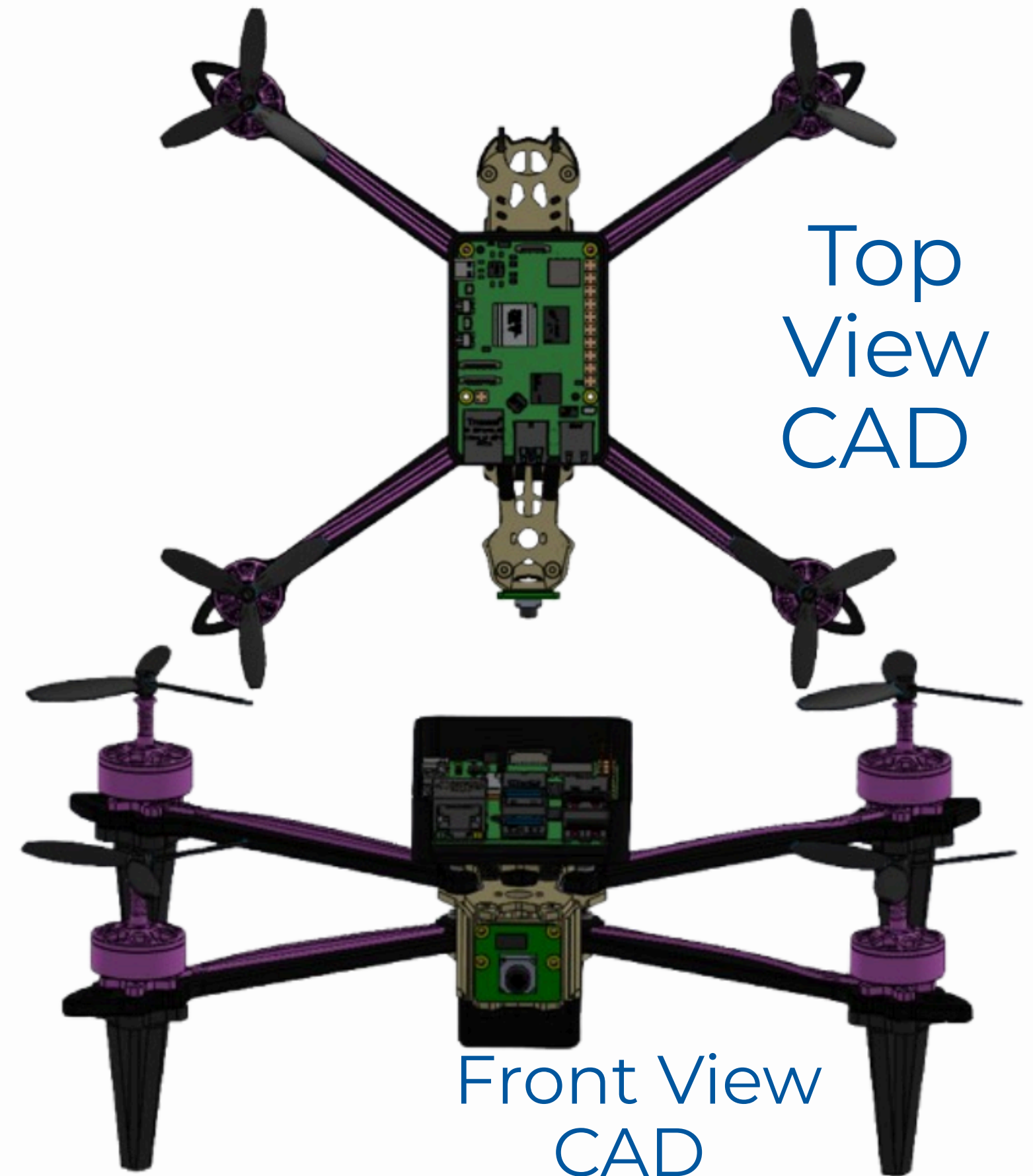
5

Follower Drone
[Human Guidance]

Status & Path Feedback

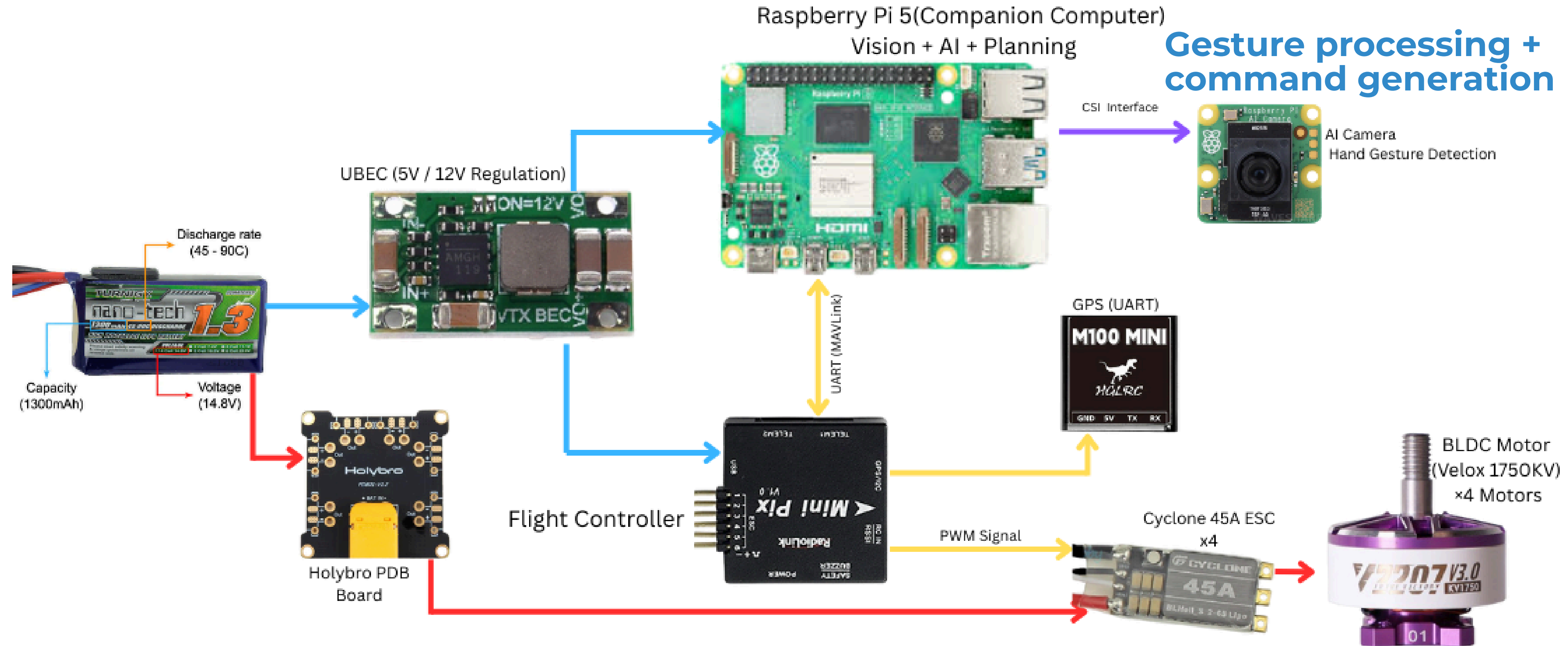
Let's Dive into the Hardware

- Custom lightweight quadrotor frame
- Velox 1750KV motors + 45A ESCs
- Mini Pix flight controller
- Raspberry Pi 5 (Leader)
- Raspberry Pi 4 (Follower drones)
- Li-Po battery system



Modular hardware design for rapid field replacement

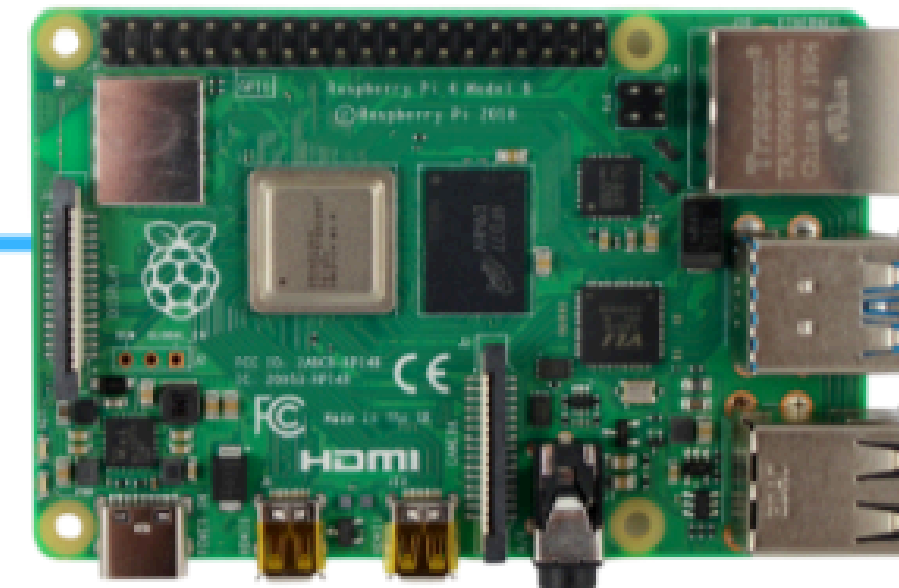
Electronics & Power Flow (Leader Drone)



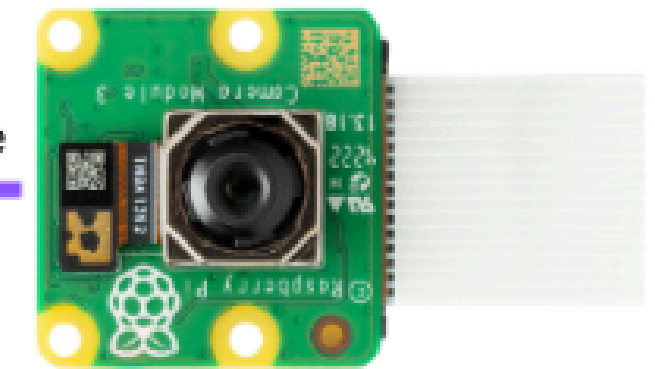
Electronics & Power Flow (Other Drone)

Vision-based detection (Detection + Execution)

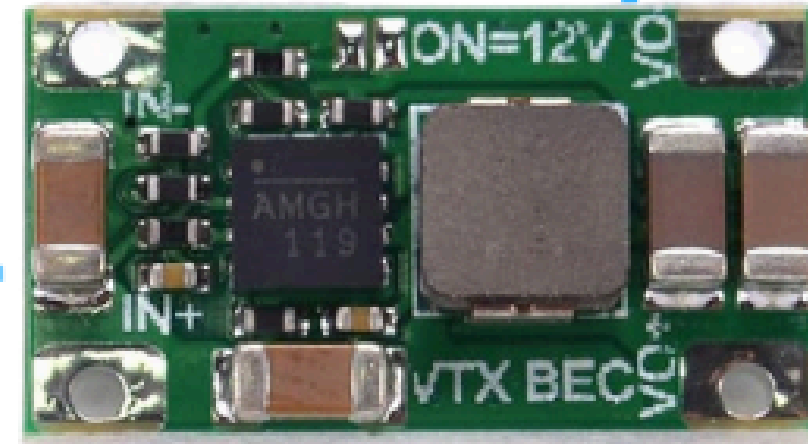
Raspberry Pi 4(Companion Computer)
Vision + Planning



CSI Interface

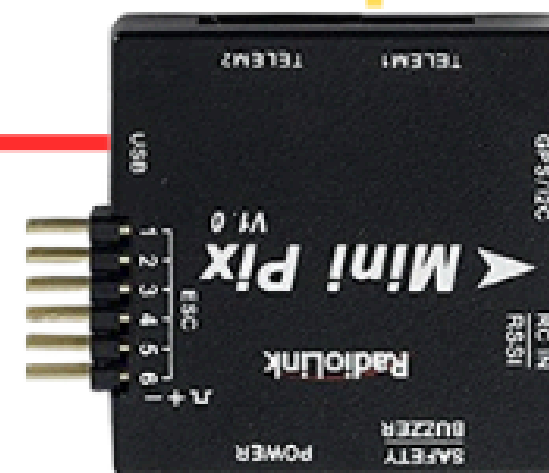


Pi Camera Module
(Mine Detection)



UBEC (5V / 12V Regulation)

UART (MAVLink)



Flight Controller

GPS (UART)

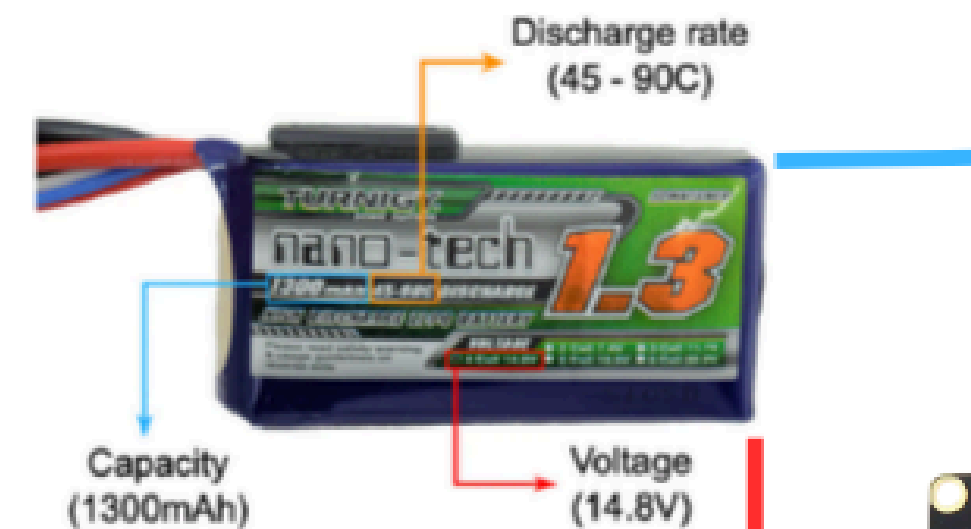


PWM Signal

Cyclone 45A ESC



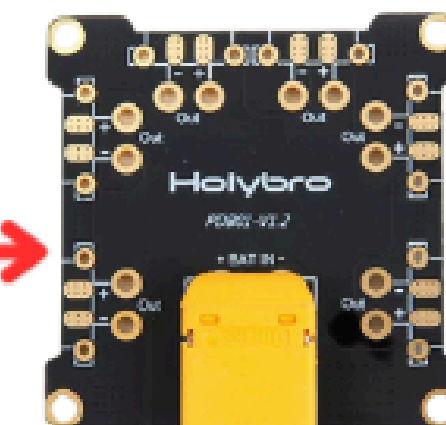
BLDC Motor
(Velox 1750KV)
×4 Motors



Discharge rate
(45 - 90C)

Capacity
(1300mAh)

Voltage
(14.8V)



Holybro PDB
Board

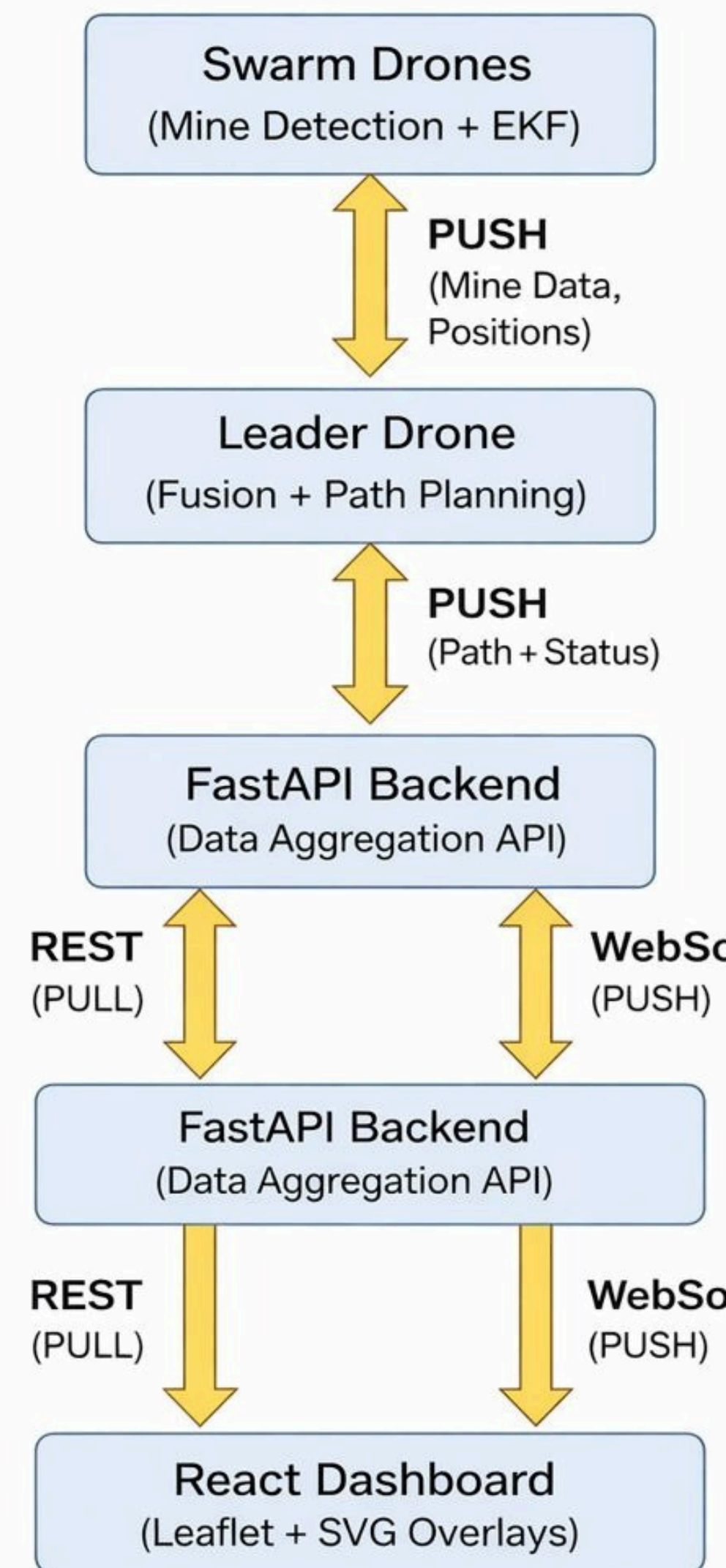
Software Stack & Data Flow

OpenCV
Vision

MediaPipe
Hand Gesture

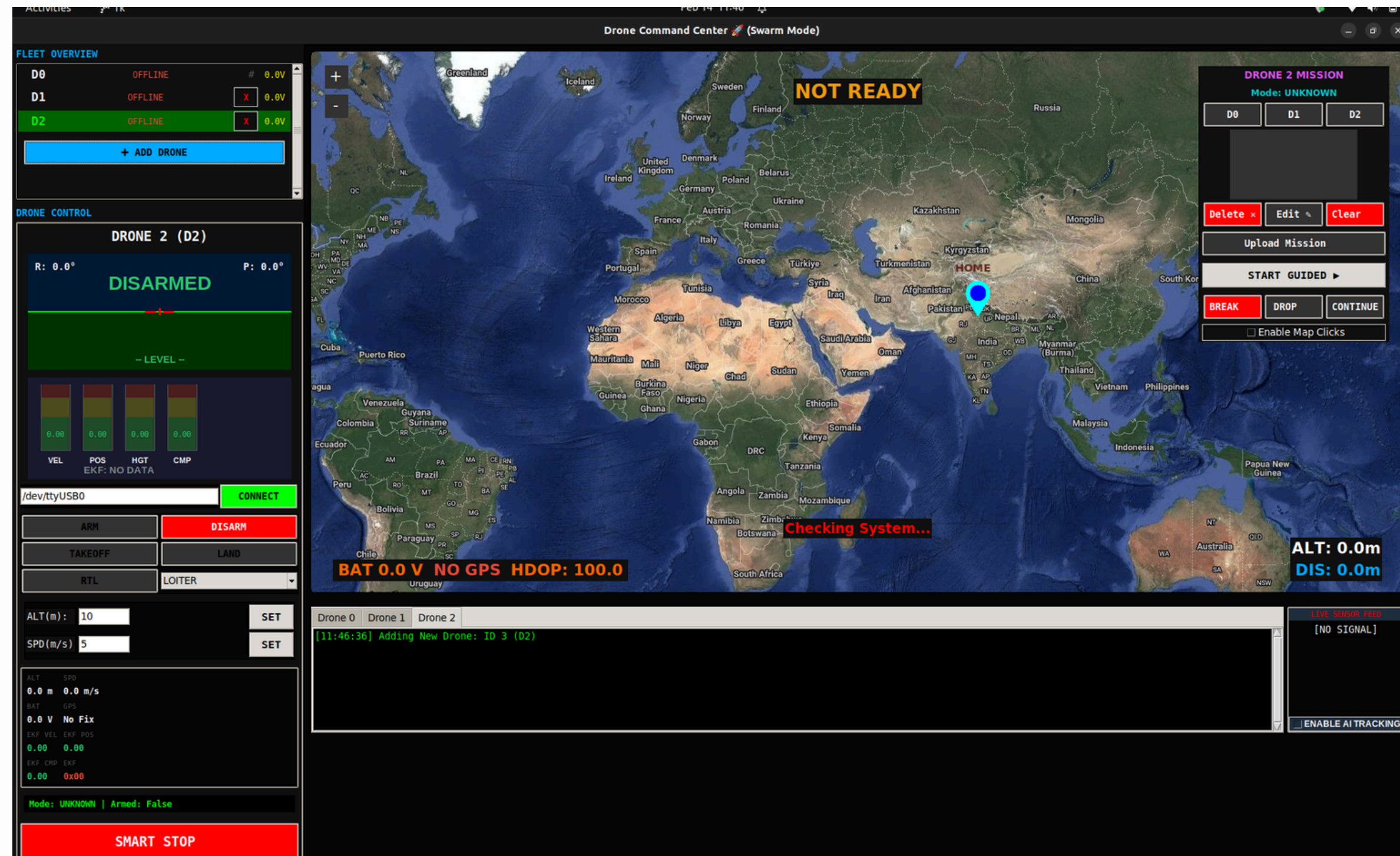
MAVLink
Flight commands

Low-latency UDP over Wi-Fi
Inter-drone comms



CUSTOM SOFTWARE

- We have designed a fully working software for swarming .
- It is a custom GCS(Ground Control System) that controls the drone and maps the coordinates in the system.
- Software Stack is fully functional and we are now working on its testing phase.



Gesture Control Demonstration

RoboFest Gujarat 5.0

Minefield Navigation Drones

Ajay Kumar Garg Engineering College



Prototype & Real Hardware



Challenges & Learnings



Power distribution &
noise isolation



Stable inter-drone
communication

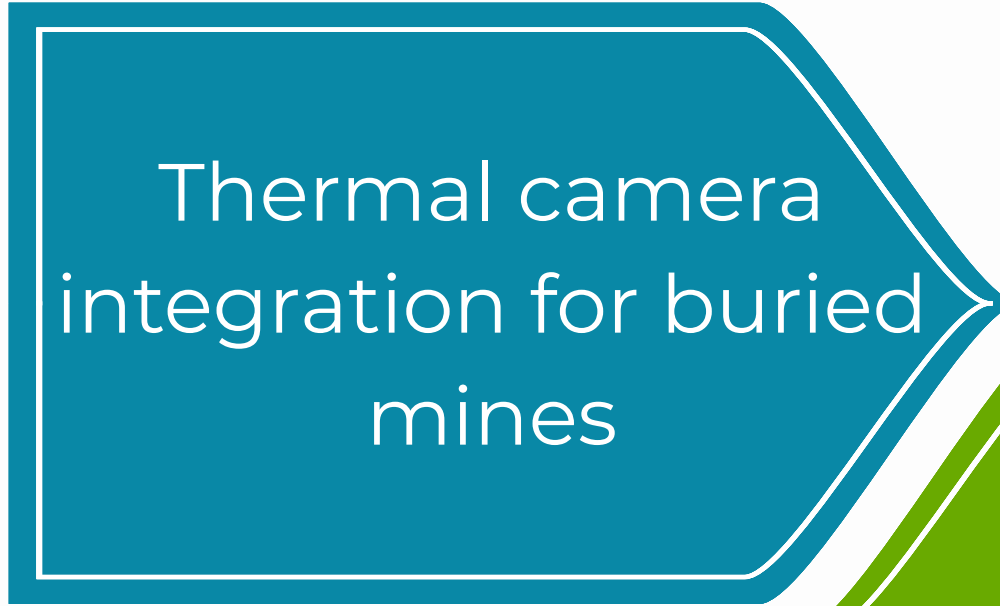


Weight VS endurance
tradeoff

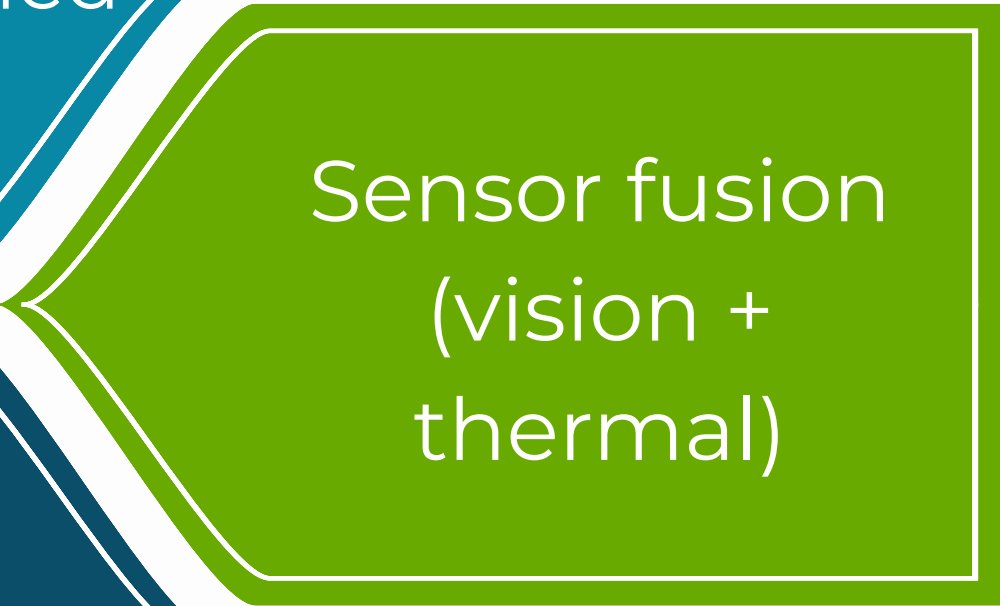


Multi-drone coordination
timing

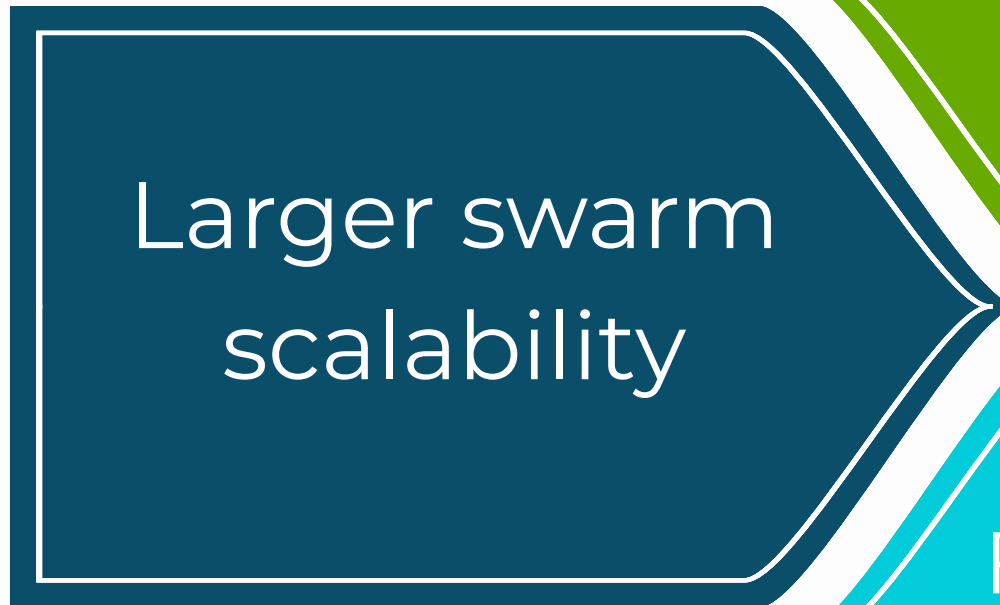
Future Scope



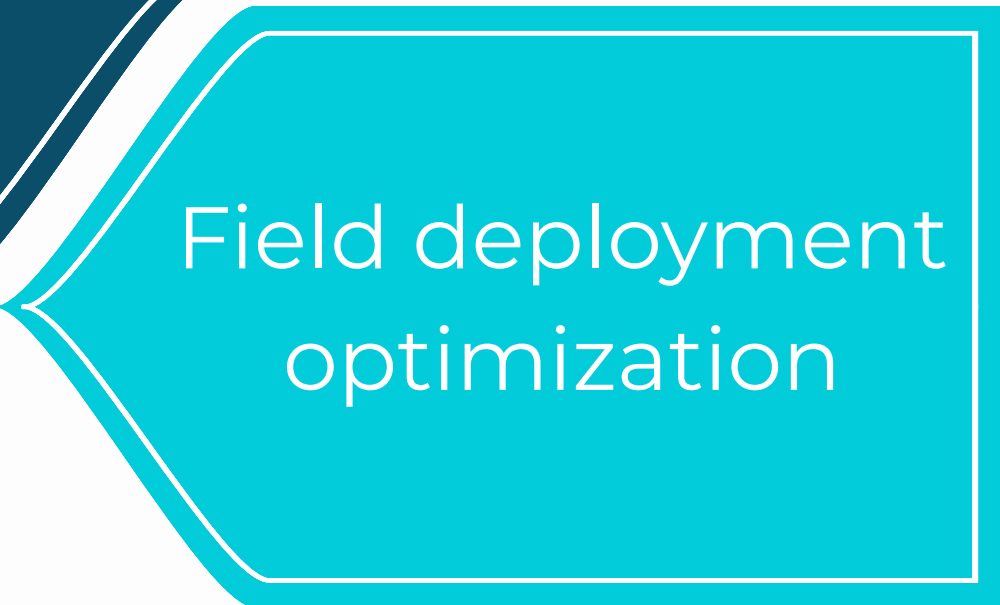
Thermal camera
integration for buried
mines



Sensor fusion
(vision +
thermal)



Larger swarm
scalability



Field deployment
optimization

Thank You